

LAB EXPERIMENT RECORD

Department	AIDS
Course Code / Course Name	DSA0601 / Data Handling and Visualization
Experiment	Data Visualization using R
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Register Number	192424085
Date	29 September 2026
Day	Tuesday

SET 3 — EMPLOYEE PERFORMANCE EVALUATION

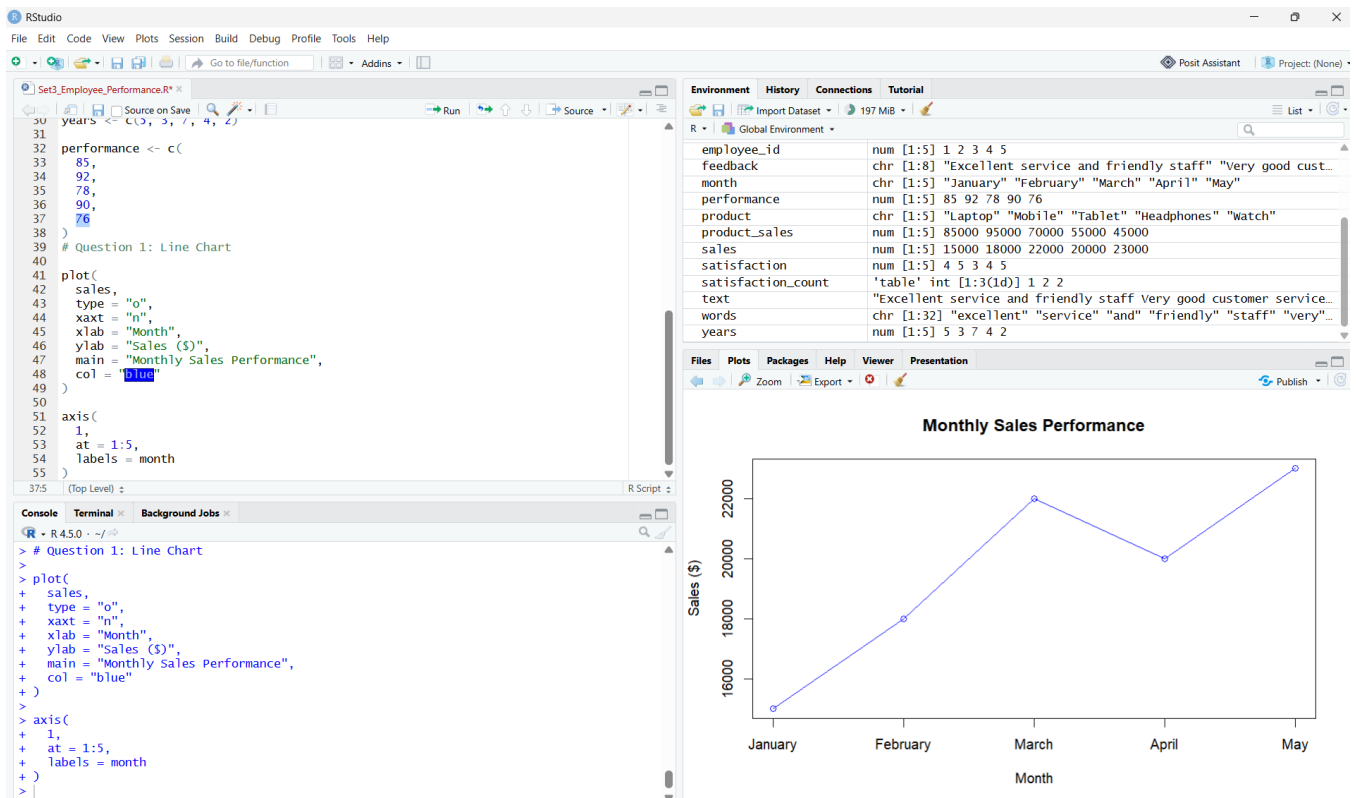
Aim:

To analyze and visualize employee performance data using line charts, bar charts, and scatter plots in R, and to develop an interactive dashboard for exploring sales trends and employee department distribution.

PROGRAM 1: LINE CHART

Create a line chart to visualize the performance trend of employees over time.

```
plot(  
  sales,  
  type = "o",  
  xaxt = "n",  
  xlab = "Month",  
  ylab = "Sales ($)",  
  main = "Monthly Sales Performance",  
  col = "blue"  
)  
axis(  
  1,  
  at = 1:5,  
  labels = month  
)
```



Output 1: Distribution of Customer Ages

Observation: Sales increased from January to May, with a small decrease in April.

PROGRAM 2: BAR CHART

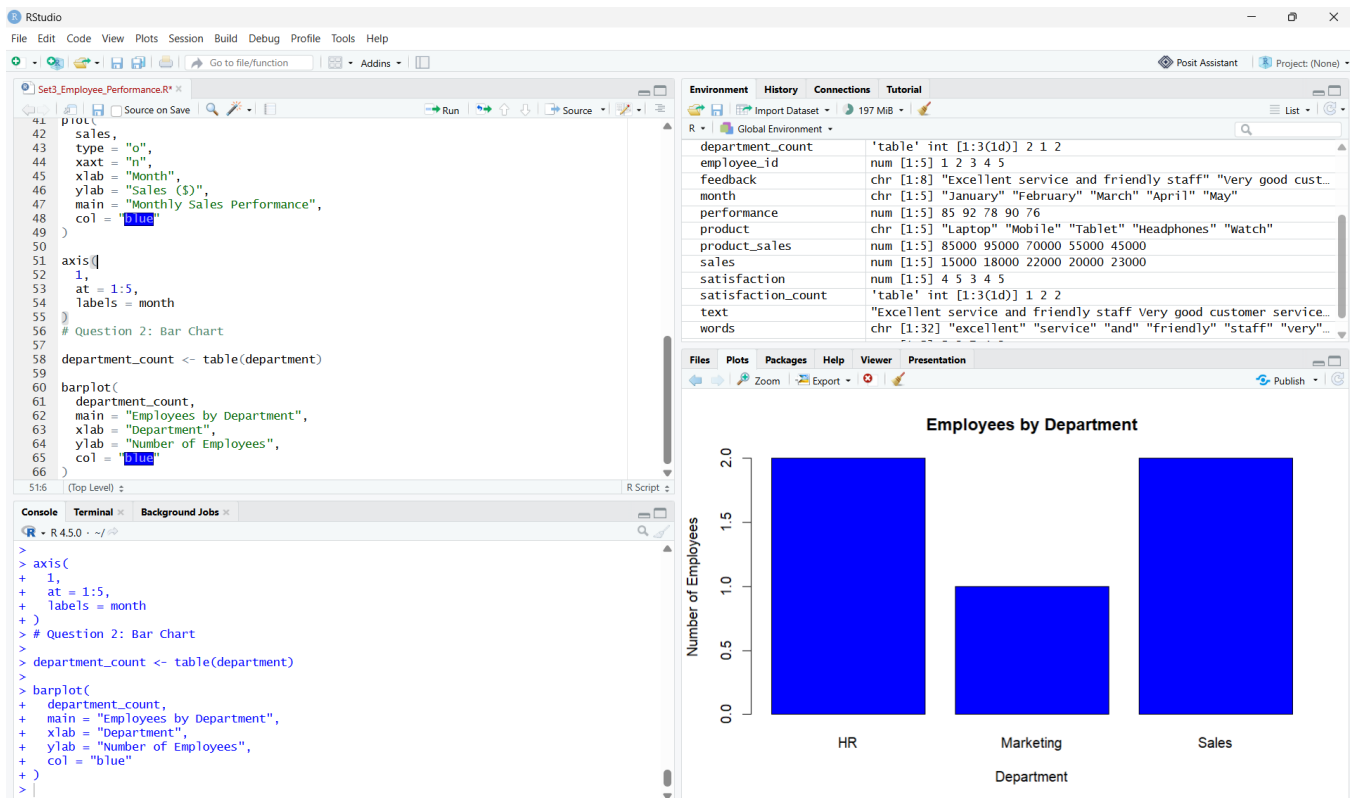
Generate a bar chart showing the distribution of employees across different departments.

```

# Question 2: Bar Chart
department_count <- table(department)

barplot(
  department_count,
  main = "Employees by Department",
  xlab = "Department",
  ylab = "Number of Employees",
  col = "blue"
)

```



Output 2: Customer Satisfaction Scores Pie Chart

Observation:

Department Employees

Sales 2

HR 2

Marketing 1

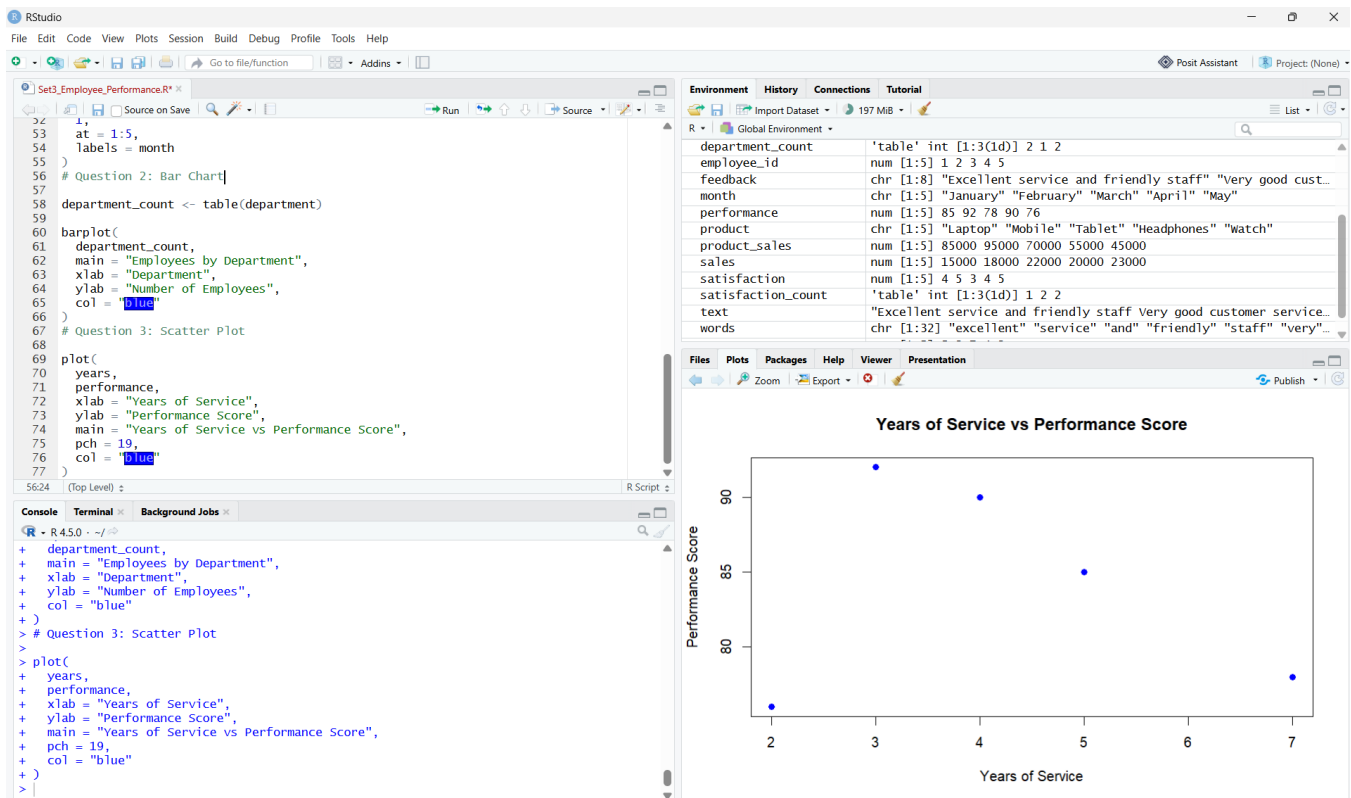
PROGRAM 3: SCATTER PLOT

Build a scatter plot to analyze the correlation between years of service and performance scores.

```

# Question 3: Scatter Plot
plot(years,
      performance,
      xlab = "Years of Service",
      ylab = "Performance Score",
      main = "Years of Service vs Performance Score",
      pch = 19,
      col = "blue")

```



Output 3: Satisfaction Scores by Age Group

Observation: The scatter plot shows the relationship between years of service and performance score. In this small dataset, the relationship is not perfectly linear, so longer service does not consistently correspond to higher performance.

PROGRAM 4: Word Cloud of Customer Feedback

Develop a Tableau dashboard with the line chart and bar chart for interactive exploration.

```

library(shiny)

# Monthly Sales Data
month <- c(
  "January",
  "February",
  "March",
  "April",
  "May")
sales <- c(
  15000,
  18000,
  22000,
  20000,
  23000)

```

```

# Employee Data
department <- c(
  "Sales",
  "HR",
  "Marketing",
  "Sales",
  "HR")

# User Interface
ui <- fluidPage(
  titlePanel(
    "Employee Performance Dashboard"),
  selectInput(
    "chart",
    "Select Chart:",
    choices = c(
      "Monthly Sales",
      "Employees by Department" ) ),
  plotOutput("myplot"))

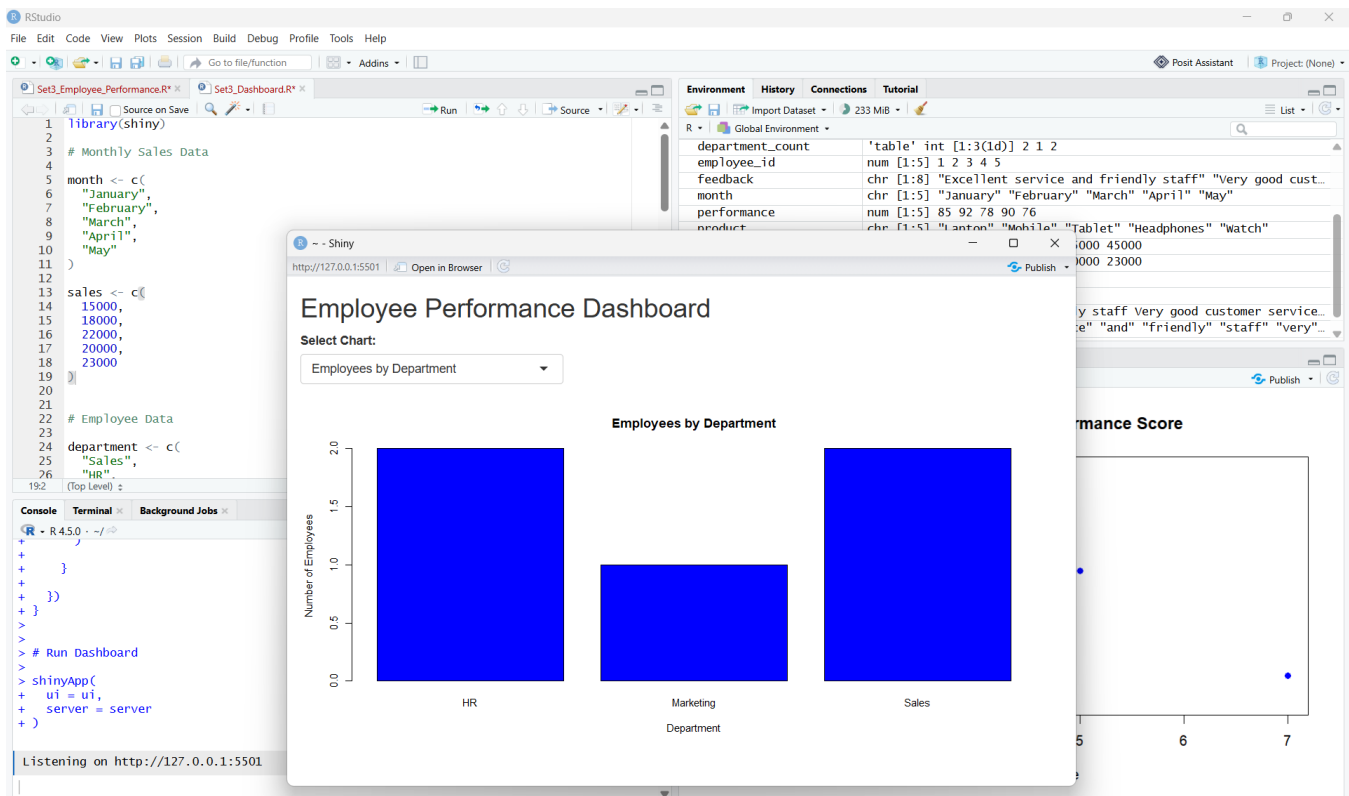
# Server
server <- function(input, output) {
  output$myplot <- renderPlot({
    if (
      input$chart == "Monthly Sales"
    ) {
      plot(
        sales,
        type = "o",
        xaxt = "n",
        xlab = "Month",
        ylab = "Sales ($)",
        main = "Monthly Sales Performance",
        col = "blue")
      axis(
        1,
        at = 1:5,
        labels = month )}
    if (
      input$chart == "Employees by Department" ) {

```

```

department_count <- table(
  department)
barplot(
  department_count,
  main = "Employees by Department",
  xlab = "Department",
  ylab = "Number of Employees",
  col = "blue" )
} } })
# Run Dashboard
shinyApp(
  ui = ui,
  server = server
)

```



Output 4: Customer Feedback Word Cloud

Observation: "The question does not provide employee performance values over different months. It provides monthly sales data, so I used the given time-based monthly dataset for the line chart."

Set 3 Procedure

1. Create vectors for monthly sales, employee department, years of service, and performance score.
2. Use `plot()` to visualize the monthly sales performance trend with month labels.

3. Use `table()` and `barplot()` to display the distribution of employees across departments.
4. Use `plot()` to create a scatter plot showing the relationship between years of service and performance scores.
5. Use `shiny()` to develop an interactive dashboard combining the monthly sales line chart and department-wise employee bar chart.

Set 3 Result

The employee performance data was successfully visualized using a line chart, bar chart, and scatter plot. An interactive dashboard was also developed to explore the monthly sales trend and employee department distribution.